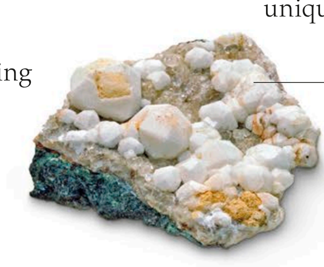
The Bay of Fundy

A deep inlet into North America's eastern seaboard, best known for its huge tides but also for its dinosaur fossils and semiprecious minerals

Located on Canada's east coast, the Bay of Fundy is a unique coastal environment with a landscape of breathtaking cliffs, sea caves, and unusual rock formations. The rock layers and outcrops around the bay display an enormous geological diversity and range of ages. They tell a story of hundreds of millions of years of natural history with, for example, dinosaur discoveries from the Triassic Period and fossils of marine invertebrates from the Carboniferous Period.

Massive tides

High cliffs channel the tidal waters flowing into the bay until they separate into two narrow arms (called Chignecto Bay and the Minas Basin) at its northeastern end. This channeling effect and the bay's overall shape give rise to some of the world's fastest-running tides and highest tidal ranges (the difference between high and low tides)—the highest ever being a range of 71 ft (21.6 m) recorded one night in October 1869 at Burntcoat Head in the Minas Basin. The tidal surge in the northern arm, Chignecto Bay, produces a bore up to 6 ft (1.8 m). The twice-daily tides transform the shorelines, tidal flats, and exposed sea bottom as they flood into the Bay and its harbors. They also create unique conditions for whales and seabirds.



◁ **MINERAL CLUSTER**
Cape Blomidon, which borders the Minas Basin, and other sites in the bay are important source areas for the white mineral analcime.

176 billion tons (160 billion tonnes) of seawater flow in and out of the bay twice a day

TIDAL RANGES

At a typical Fundy location like Cape Blomidon (see graph below), the average tidal range is about 39 ft (12 m). But at some locations, it can be as high as 65 ft (20 m). As well as depending on location, tidal ranges are affected by the monthly lunar cycle.